Team Work Solution Company

Guesthouse Reservation Platform

A Self-Service Reservation & Property-Operations Platform for the Ethiopian Hospitality Market

Revision Note

This revision (Version 2.0) restructures the platform around a simplified four-role model — Guest, Guesthouse Owner, Receptionist, and System Administrator — supported by an automated System layer that handles authentication, availability, online payment verification, and notifications. The Cashier/Accountant and Supervisor roles, and the manual cash-handling and housekeeping workflows from Version 1.0, are removed in favor of automated equivalents.

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1. Introduction

1.1 Purpose of this Document

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Guesthouse Reservation Platform. It describes the system's scope, the roles that operate it, the workflows each role follows, the data model, the technology stack, and the quality and security standards the platform must meet. This document supersedes the Version 1.0 draft, which modeled the platform around six roles including a Cashier/Accountant and a Supervisor. This revision reflects a leaner operating model in which reservations are self-service and online-payment-only, and in which the System itself performs authentication, availability management, payment verification, and notifications automatically.

1.2 Project Overview

The Guesthouse Reservation Platform is a modern, full-stack web application designed to streamline the process of searching, booking, and managing guesthouse reservations for the Ethiopian hospitality market. It connects travelers seeking accommodation with guesthouse businesses, and it gives each guesthouse's Owner and front-desk Receptionist dedicated tools to run daily operations, while the platform's automated System handles booking creation, availability, payment verification, and guest communication.

In Ethiopia, many guesthouses still rely on manual, phone-based reservation handling, which leads to missed bookings, double bookings, and poor visibility for travelers. Small and mid-sized guesthouses typically lack dedicated software to manage reservations, front-desk operations, and payments in an organized, auditable way. This platform is built specifically for that market, with a self-service booking flow and automated safeguards, and with future support for local payment methods such as Telebirr and Chapa.

1.3 Problem Statement

Traditional guesthouse reservation systems suffer from a combination of technical and operational gaps:

- Customers cannot easily compare guesthouses, prices, or availability online.
- Room availability is not updated in real time, causing double bookings.
- Owners and staff spend significant time managing reservations manually by phone or paper log.
- Payments collected outside the system are not tracked consistently, making reconciliation and financial reporting difficult.
- Owners have no reliable, centralized way to monitor staff activity, occupancy, or revenue across their property.

The platform addresses these gaps by making reservation creation, availability management, and payment verification automatic functions of the System itself, while giving the Owner, Receptionist, and Admin their own accountable, role-scoped tools.

1.4 Objectives

The overarching aim of this project is to develop a secure, user-friendly, and operationally realistic web-based Guesthouse Reservation Platform that improves the efficiency of reservation management and daily property operations. Specific objectives:

- Allow guests to search, compare, and book guesthouse rooms online, and pay online.
- Prevent double booking through automatic, transaction-safe availability checks performed by the System.
- Give front-desk staff (Receptionist) the tools to manage arrivals, departures, check-in, check-out, and room availability.

- Give guesthouse Owners the tools to manage their property, rooms, and staff, and to view payment history and revenue reports.
- Give the System Administrator the tools to approve guesthouse registrations, manage accounts, maintain system settings, and monitor platform activity.
- Let the System automatically authenticate and authorize users, create and manage reservations, verify online payments, record transactions, generate receipts, and send confirmation notifications.
- Provide secure authentication and strict role-based access control across all roles.
- Deploy a responsive, reliable application suitable for both desktop front-desk use and mobile guest use.

2. Project Scope

2.1 In Scope

- User registration and authentication for all roles (Guest, Guesthouse Owner, Receptionist, System Administrator).
- Guest profile management, guesthouse search, and viewing of guesthouse and room details.
- Automatic, real-time room availability checking, handled by the System.
- Online room booking and online payment by the Guest.
- Automatic reservation creation and automatic reservation-status management by the System.
- Automatic prevention of double booking by the System.
- Automatic verification of online payments, recording of transactions, receipt generation, and storage of payment history.
- Automatic booking-confirmation and payment-confirmation notifications to the Guest.
- Reservation history for the Guest.
- Receptionist dashboard: reservation list, today's arrivals, today's departures.
- Guest check-in and check-out performed by the Receptionist.
- Room availability updates (Available / Unavailable) performed by the Receptionist.
- Owner tools: guesthouse registration, guesthouse management, room management, staff management, payment history, and revenue reports.
- Admin tools: guesthouse-registration approval, guesthouse-account management, user-account management, system-settings management, platform reporting, and system-activity monitoring.
- Role-based access control (RBAC) enforced on every module and API endpoint.
- Secure online payment processing, with future integration of Telebirr and Chapa.

2.2 Out of Scope

- Cash or in-person (walk-in/phone) payment recording — all payments are processed online by the System.
- Manual (walk-in/phone) reservation entry by staff — reservations are created automatically by the System from Guest bookings.
- Housekeeping / room-cleaning lifecycle tracking (dirty, cleaning, inspected, maintenance) — room status is limited to Available / Unavailable.
- Guest reviews and ratings (planned for a future phase).
- Flight booking
- Hotel chain / large multi-brand hospitality management
- Restaurant reservations

- Vehicle rentals
- Travel insurance
- Tour guide booking
- Native mobile application (planned for Phase 2)
- SMS notifications (planned for Phase 2)
- AI-powered recommendations (planned for Phase 3)

3. Stakeholders and System Roles

3.1 Role Hierarchy

The platform is used by one external stakeholder (the Guest) and three internal human roles, supported throughout by an automated System layer that performs functions no human role needs to trigger manually. The internal roles are organized in two tiers: a platform tier (System Administrator) that operates across all guesthouses, and a property tier (Guesthouse Owner, Receptionist) that operates within a single guesthouse.

Platform Tier

- System Administrator — oversees the entire platform across all guesthouses: approves new guesthouse registrations, manages guesthouse and user accounts, maintains system settings, monitors platform activity, and produces platform-wide reports.

Property Tier (per guesthouse)

- Guesthouse Owner — top of the property's hierarchy. Registers and manages the guesthouse, manages rooms, manages staff accounts, and reviews payment history and revenue reports.
- Receptionist — reports to the Owner. Runs day-to-day front-desk operations: viewing the dashboard, the reservation list, and today's arrivals and departures; performing guest check-in and check-out; and updating room availability.

Automated Tier

- System (Automatic Functions) — not a human role, but the engine underlying every other role. It authenticates and authorizes users, checks room availability, creates reservations, manages reservation status and room availability automatically, prevents double booking, verifies online payments, records payment transactions, generates payment receipts, stores payment history, and sends booking- and payment-confirmation notifications.

A booking made by a Guest is created directly by the System once payment is verified — there is no manual confirmation step. The Receptionist manages the guest's arrival and departure and keeps room availability current; the Owner can see all activity for their property; and the System Administrator is responsible for the platform infrastructure and accounts that make all of this possible.

3.2 Role Descriptions and Responsibilities

Role	Main Responsibilities	Reports To
Guest (Customer)	Registers/logs in, searches guesthouses, views guesthouse and room details, books and pays for rooms online, views reservation history, updates profile, logs out.	N/A (external user)
Guesthouse Owner	Registers and manages the guesthouse, manages rooms,	Platform (self-managed

Role	Main Responsibilities	Reports To
	manages staff accounts, views payment history and revenue reports, updates profile, logs out.	business)
Receptionist	Views dashboard, reservation list, and today's arrivals/departures; performs guest check-in and check-out; updates room availability (Available/Unavailable); logs out.	Guesthouse Owner
System Administrator	Approves guesthouse registrations, manages guesthouse and user accounts, manages system settings, views platform reports, monitors system activity, updates profile, logs out.	Platform (technical operator)
System (Automatic Functions)	Authenticates and authorizes users; checks availability; creates reservations; manages reservation status and room availability automatically; prevents double booking; verifies online payments; records transactions; generates receipts; stores payment history; sends booking and payment confirmation notifications.	N/A (platform engine)

4. User Journeys and Workflows

4.1 Guest Booking Workflow

- User lands on the homepage and views featured guesthouses and the search bar.
- User searches guesthouses and opens a listing to view its details, room types, and pricing.
- User views available rooms for their chosen dates.
- User selects a room and proceeds to book it.
- User makes an online payment for the booking.
- The System verifies the payment, automatically creates the reservation, records the transaction, generates a receipt, and sends booking- and payment-confirmation notifications.
- User can later view their reservation history and update their profile.

4.2 Receptionist Front-Desk Workflow

- Receptionist logs in and views the dashboard.
- Receptionist views the reservation list and today's arrivals and departures.
- At arrival, the Receptionist performs guest check-in.
- At departure, the Receptionist performs guest check-out.
- The Receptionist updates room availability (Available / Unavailable) as needed, for example after check-out or when a room is taken out of service.

4.3 Guesthouse Owner Property-Management Workflow

- Registers the guesthouse and submits it for platform approval.
- Manages the guesthouse profile, rooms, and pricing.
- Creates and manages Receptionist staff accounts for the property.
- Views payment history for the property.

- Views revenue reports for the property.
- Updates their own profile.

4.4 System Administrator Platform-Oversight Workflow

- Reviews pending guesthouse registrations and approves them.
- Manages guesthouse accounts across the platform.
- Manages user accounts across the platform.
- Manages system settings.
- Views platform-wide reports.
- Monitors system activity.

4.5 System Automated Workflow

The following functions run automatically, without a human role initiating them directly:

- Authenticating users and authorizing their actions based on role.
- Checking room availability at the moment of booking.
- Creating the reservation once payment is verified.
- Managing reservation status (e.g., confirmed, checked-in, checked-out, cancelled) as the booking progresses.
- Managing room availability as reservations are created, checked out, or cancelled.
- Preventing double booking through transaction-safe checks.
- Verifying online payments and recording the transaction.
- Generating a payment receipt and storing payment history.
- Sending booking-confirmation and payment-confirmation notifications to the Guest.

5. Business Rules

5.1 Reservation Rules

- A room cannot be booked twice for overlapping dates; the System enforces this automatically at booking time.
- Check-out date must be after check-in date; minimum stay duration is 1 night.
- Guests can only book rooms currently marked Available.
- A reservation is created by the System only after the corresponding online payment has been verified.
- Check-in and check-out may only be performed by the Receptionist (or the Owner) for the guesthouse the reservation belongs to.

5.2 Guesthouse and Staffing Rules

- Only approved guesthouses appear in public search results.
- Only a verified Owner can register a guesthouse and add rooms; the guesthouse must have at least one room to be listed.
- Only the Owner (or the Admin, for support purposes) can create Receptionist accounts for that property.
- Receptionist accounts are scoped to a single guesthouse and cannot access another property's data.
- Owners can update room details and pricing in real time.

5.3 Payment Rules

- All payments are made online by the Guest; the platform does not record cash or in-person payments.
- The System verifies every online payment before the reservation is created.
- The System automatically records the payment method, amount, and reservation reference, and generates a receipt.
- Failed online payments may be retried up to 3 times before the room is released back to availability.
- Every recorded payment is linked to exactly one reservation for audit purposes.
- Future support for Telebirr and Chapa payments will follow the same automated verification and recording rules as other online payment methods.

5.4 Room-Availability Rules

- Each room has an availability status of either Available or Unavailable.
- The System automatically sets a room to Unavailable once it is successfully booked for the relevant dates.
- The Receptionist may manually mark a room Available or Unavailable, for example after check-out or when a room must be taken out of service.

5.5 Administrative Rules

- Only the System Administrator can approve new guesthouses onto the platform.
- Only the System Administrator can suspend or delete accounts for cause (e.g., fraud) across the platform.
- The System Administrator can view platform-wide operational data but does not alter a specific guesthouse's day-to-day rooms, reservations, or payment records — those remain under the Owner's authority.

6. Functional Requirements

Functional Requirements describe what the system must do. They define all the functions and services that the system provides to different users.

6.1 Shared / Authentication Requirements

- The system shall allow user registration, login, and logout for all roles.
- The system shall issue a signed token (JWT) on login and require it for all protected actions.
- The system shall enforce role-based authorization on every endpoint, scoped to the user's guesthouse where applicable.

6.2 Guest Requirements

- The system shall allow guests to register an account, log in, and log out.
- The system shall allow guests to search guesthouses.
- The system shall allow guests to view guesthouse details.
- The system shall allow guests to view available rooms.
- The system shall allow guests to book a room.
- The system shall allow guests to make an online payment for a booking.
- The system shall allow guests to view their reservation history.
- The system shall allow guests to update their profile.

6.3 Guesthouse Owner Requirements

- The system shall allow the Owner to register a guesthouse.
- The system shall allow the Owner to manage the guesthouse profile.
- The system shall allow the Owner to manage rooms.
- The system shall allow the Owner to manage staff (Receptionist) accounts.
- The system shall allow the Owner to view payment history for their property.
- The system shall allow the Owner to view revenue reports for their property.
- The system shall allow the Owner to update their profile.

6.4 Receptionist Requirements

- The system shall allow the Receptionist to log in and log out.
- The system shall display a dashboard to the Receptionist.
- The system shall display a reservation list to the Receptionist.
- The system shall display today's arrivals to the Receptionist.
- The system shall display today's departures to the Receptionist.
- The system shall allow the Receptionist to perform guest check-in.
- The system shall allow the Receptionist to perform guest check-out.
- The system shall allow the Receptionist to update room availability (Available/Unavailable).

6.5 System Administrator Requirements

- The system shall allow the Admin to log in and log out.
- The system shall allow the Admin to approve guesthouse registrations.
- The system shall allow the Admin to manage guesthouse accounts.
- The system shall allow the Admin to manage user accounts.
- The system shall allow the Admin to manage system settings.
- The system shall allow the Admin to view platform reports.
- The system shall allow the Admin to monitor system activities.
- The system shall allow the Admin to update their profile.

6.6 System (Automatic Function) Requirements

- The system shall automatically authenticate users.
- The system shall automatically authorize users based on their role.
- The system shall automatically check room availability at the time of booking.
- The system shall automatically create reservations once payment is verified.
- The system shall automatically manage reservation status as a booking progresses.
- The system shall automatically manage room availability as bookings, check-outs, and cancellations occur.
- The system shall automatically prevent double booking.
- The system shall automatically verify online payments.
- The system shall automatically record payment transactions.
- The system shall automatically generate payment receipts.
- The system shall automatically store payment history.

- The system shall automatically send booking-confirmation notifications.
- The system shall automatically send payment-confirmation notifications.

7. Non-Functional Requirements

Non-Functional Requirements describe how the system works, not what the system does. They focus on the system's quality, speed, security, reliability, and performance.

7.1 Performance

- Search results shall appear within two seconds under normal load.
- The system shall handle concurrent users efficiently, including simultaneous front-desk and guest activity on the same property.
- Database queries shall be optimized for speed, particularly availability checks.

7.2 Security

- Passwords shall be encrypted using bcrypt; plain-text storage is prohibited.
- JWT authentication shall be used, with session/token expiration.
- All protected routes shall require authentication and shall enforce role-based access control.
- Data shall be encrypted in transit (HTTPS/TLS) and at rest.
- Every payment and account-management action shall be logged with the acting user (or the System, for automated actions) for audit purposes.

7.3 Reliability

- Reservation transactions shall use database-level locking to prevent duplicate bookings.
- The system shall target 99.5% uptime.
- Automated daily backup and recovery procedures shall be in place.

7.4 Usability

- The interface shall be mobile-responsive for guests and clear/efficient for staff workflows on desktop devices at the front desk.
- Each role shall see a dashboard scoped to its own responsibilities, not a generic one-size-fits-all screen.
- The system shall follow accessibility best practices and support easy navigation.

8. System Architecture

The Guesthouse Reservation Platform is built on a three-tier architecture, ensuring scalability, maintainability, and a clear separation of concerns. This layered approach allows independent development, deployment, and scaling of each component.

8.1 Presentation Layer (Frontend)

- React.js user interface, with distinct views/dashboards per role.
- Responsive design using Tailwind CSS.
- Client-side routing with React Router, gated by role-based route guards.

- HTTP requests via Axios.

8.2 Application Layer (Backend)

- Node.js runtime environment.
- Express.js web framework.
- RESTful API endpoints, each protected by authentication and role-based authorization middleware.
- Business logic for automatic reservation creation, availability management, payment verification, and notifications.

8.3 Data Layer (Database)

- PostgreSQL relational database.
- Prisma ORM for type-safe database operations and migrations.
- Transaction support to guarantee reservation and payment consistency.

8.4 Role-Based Access Control Layer

Cutting across all three tiers, an RBAC layer maps each of the four human roles to a defined set of permitted actions and data scopes (platform-wide for the Admin; property-scoped for the Owner and Receptionist; self-scoped for Guests), while the automated System layer executes availability, payment, and notification logic on behalf of the platform as a whole. This is enforced both in the UI (hiding unavailable actions) and in the API (rejecting unauthorized requests).

9. Database Design

The database is structured to efficiently manage users and staff, guesthouses, rooms and their availability, reservations, and payments. The Users table supports four human roles. Room status is simplified to Available/Unavailable, and the Payments table reflects online-only, System-verified transactions.

9.1 Users Table

Field	Description
user_id	Primary Key
email, password (hashed), name, phone	Core account and contact fields
role	One of: Guest, Owner, Receptionist, Admin
guesthouse_id	Foreign Key, nullable — set for property-scoped staff (Owner, Receptionist); null for Guest and platform Admin
created_at, updated_at	Record timestamps

9.2 Guesthouses Table

Field	Description
guesthouse_id	Primary Key
owner_id	Foreign Key → Users

Field	Description
name, description, location, address	Property profile details
status	pending / approved / rejected
created_at, updated_at	Record timestamps

9.3 Rooms Table

Field	Description
room_id	Primary Key
guesthouse_id	Foreign Key → Guesthouses
room_number, type, capacity	Room identification and capacity
price_per_night	Nightly rate
availability_status	available / unavailable
created_at, updated_at	Record timestamps

9.4 Reservations Table

Field	Description
reservation_id	Primary Key
guest_id	Foreign Key → Users
room_id	Foreign Key → Rooms
created_by	System — reservations are created automatically once payment is verified
check_in_date, check_out_date	Stay dates
status	confirmed / checked_in / checked_out / cancelled
total_price, checked_in_at, checked_out_at	Financial and timing detail
created_at, updated_at	Record timestamps

9.5 Payments Table

Field	Description
payment_id	Primary Key
reservation_id	Foreign Key → Reservations
processed_by	System — all payments are processed and verified automatically
amount, method	method: online / card / mobile-money

Field	Description
status	pending / completed / failed
created_at	Record timestamp

10. Modules and Features

The platform is structured around interconnected modules, each aligned to one of the four human roles, plus the automated System module. This modular approach ensures clarity, scalability, and efficient development.

10.1 Authentication Module

- User registration, login, logout
- JWT token management and role-based route protection

10.2 Guest Module

- Search guesthouses
- View guesthouse details and available rooms
- Book a room and pay online
- View reservation history
- Profile management

10.3 Receptionist Module

- Dashboard, reservation list, today's arrivals and departures
- Guest check-in and check-out
- Room availability updates (Available/Unavailable)

10.4 Guesthouse Owner Module

- Guesthouse registration and management
- Room management
- Staff (Receptionist) account management
- Payment history and revenue reports
- Profile management

10.5 System Administrator Module

- Guesthouse-registration approval
- Guesthouse-account management
- User-account management
- System-settings management
- Platform reporting and system-activity monitoring

10.6 System (Automatic Functions) Module

- Authentication and role-based authorization
- Room availability checks and double-booking prevention
- Automatic reservation creation and status management
- Online payment verification, transaction recording, and receipt generation
- Payment-history storage
- Booking- and payment-confirmation notifications

11. Technology Stack

Our platform is built upon a robust and modern technology stack, ensuring scalability, security, and a seamless user experience across all functionalities.

Layer	Technology	Role
Frontend	React.js	Building dynamic, role-aware user interfaces
	Tailwind CSS	Utility-first styling for rapid UI development
	React Router	Client-side routing and role-based route guarding
	Axios	Asynchronous HTTP requests to the backend API
Backend	Node.js	Server-side JavaScript runtime
	Express.js	REST API and route/middleware framework
Database	PostgreSQL	Relational database for reliability and data integrity
	Prisma ORM	Type-safe queries and migrations
Authentication & Security	JWT	Stateless authentication
	bcrypt	Secure password hashing
Development & Deployment	Git & GitHub	Version control and collaboration
	Postman	API testing and documentation
	VS Code	Development environment
	Render	Backend deployment (TLS, global CDN, DDoS protection, auto-deploy from Git)
	Netlify	Frontend hosting and deployment

12. API Specification (Representative Endpoints)

All endpoints below require a valid JWT and enforce the role-based access control rules from Section 3. Endpoints marked with a role in parentheses are restricted to that role (or to the Owner and Admin, who may act within their scope). Endpoints marked (System) run automatically and are not directly invoked by a human role.

12.1 Authentication Endpoints

Endpoint	Description	Response
POST /api/auth/register	Register a new account (role assigned by context)	201 Created
POST /api/auth/login	Authenticate and receive a JWT	200 OK
POST /api/auth/logout	Invalidate the current session	200 OK

12.2 Guesthouse and Room Endpoints

Endpoint	Description	Response
GET /api/guesthouses	Search guesthouses (?location=&price_min=&price_max=)	200 OK
GET /api/guesthouses/:id	Full guesthouse detail and rooms	200 OK
POST /api/guesthouses (Owner)	Register a new guesthouse for approval	201 Created
GET /api/guesthouses/:id/rooms	View available rooms for a guesthouse	200 OK
PUT /api/rooms/:id/availability (Receptionist)	Update a room's availability status	200 OK

12.3 Reservation Endpoints

Endpoint	Description	Response
POST /api/reservations (Guest)	Initiate a booking; the System creates the reservation once payment is verified	201 Created
GET /api/reservations/:id	Retrieve full reservation detail	200 OK
GET /api/reservations (Guest)	View reservation history	200 OK
GET /api/receptionist/arrivals	Today's arrivals (Receptionist)	200 OK
GET /api/receptionist/departures	Today's departures (Receptionist)	200 OK
PUT /api/reservations/:id/checkin (Receptionist)	Perform guest check-in	200 OK
PUT /api/reservations/:id/checkout (Receptionist)	Perform guest check-out	200 OK

12.4 Payment Endpoints (System)

Endpoint	Description	Response
POST /api/payments/initiate (Guest)	Start an online payment for a reservation	201 Created
POST /api/payments/webhook (System)	Receive and verify payment-gateway confirmation	200 OK
GET /api/payments/history (Owner)	View payment history for the property	200 OK
GET /api/reports/revenue (Owner)	Revenue reports for the property	200 OK

12.5 Administration Endpoints

Endpoint	Description	Response
PUT /api/admin/guesthouses/:id/approve (Admin)	Approve a pending guesthouse registration	200 OK
GET /api/admin/users (Admin)	Manage user accounts across the platform	200 OK
GET /api/admin/reports (Admin)	Platform-wide reports	200 OK
GET /api/admin/activity (Admin)	Monitor system activity	200 OK

13. Risk Analysis

Risk	Impact	Mitigation Strategy
Server downtime	Users and staff cannot access the platform; lost bookings	Daily automated backups, redundant servers, 99.5% uptime SLA, monitoring alerts
Double booking	Customer dissatisfaction, revenue loss, reputation damage	Database transaction locks, date validation, real-time availability updates, automated conflict detection
Failed or unverified online payment	Guest is charged without a confirmed reservation, or a room is held without payment	Automated payment verification before reservation creation, limited retry attempts, automatic release of held rooms
Weak passwords	Account compromise, unauthorized access, data breach	Password policy enforcement, bcrypt hashing, password strength meter, optional 2FA
Unauthorized access / privilege escalation	Data theft, privacy violation, cross-property data exposure	JWT authentication, strict role-based access control, protected endpoints, regular security audits
Payment fraud	Financial loss, legal issues, customer trust loss	PCI DSS-aligned handling, SSL/TLS encryption, fraud detection, payment gateway validation
Data loss	Loss of critical business and guest data	Daily backups, disaster recovery plan, database replication, version control
SQL injection	Database compromise, data theft	Parameterized queries, input validation, ORM usage (Prisma), security testing
DDoS attacks	Service unavailability, reputation damage	Rate limiting, CDN protection, firewall rules, traffic monitoring
Scalability issues	Slow performance during peak times, poor user experience	Load balancing, database optimization, caching, horizontal scaling
Third-party payment-gateway failure	Payment processing failure, service interruption	Fallback payment methods, service monitoring, vendor SLA agreements

14. Future Roadmap

Our development plan is structured in distinct phases, ensuring continuous improvement and expansion of features to meet evolving user needs and market demands.

Version 2.0 — Current (MVP)

Q1–Q2 2026 | Status: In Development

- User registration and authentication for all four roles
- Guest search, viewing, online booking, and online payment
- System-automated reservation creation, availability management, and double-booking prevention
- System-automated payment verification, recording, and receipt generation
- Receptionist dashboard, check-in, check-out, and availability updates
- Owner property, room, and staff management, plus payment history and revenue reports
- Admin guesthouse-approval and account-management workflow

Version 3.0 — Phase 2

Q3–Q4 2026 | Status: Planned

- Email and SMS notifications for bookings and updates
- Additional payment gateway integrations (Telebirr, Chapa)
- Advanced search filters (amenities, distance)
- Guest reviews and ratings
- Advanced owner analytics dashboard
- Multi-language support (Amharic, English)
- Mobile-responsive design improvements for front-desk tablets

Version 4.0 — Phase 3

Q1–Q2 2027 | Status: Planned

- Native mobile app (iOS & Android) for guests and staff
- AI-powered guesthouse recommendations
- Google Maps integration
- Advanced analytics and reporting across properties
- Loyalty program for frequent guests
- Multi-property management for owners with several guesthouses

Version 5.0 — Phase 4

Q3–Q4 2027 | Status: Future Consideration

- API for third-party integrations
- White-label solution for tourism companies
- Advanced fraud detection
- Multi-currency support
- Custom reporting tools
- Integration with accounting software

15. Expected Outcomes

Our platform is designed to deliver significant improvements across various aspects of guesthouse management and guest satisfaction:

Outcome	Description
24/7 Online Availability	Guests can browse and book at any time, not only during phone hours.
Fully Self-Service Booking	Automating reservation creation and payment verification removes manual handling and reduces administrative burden.
No Double Bookings	Automated, transaction-safe availability management protects inventory accuracy.
Reliable, Automated Payment Handling	System-verified online payments and automatic receipt generation remove the need for manual cash reconciliation.
Better Customer Experience	A seamless, intuitive flow from search to booking to automatic confirmation.
Operational Efficiency for Owners	Centralized room, staff, and payment-history management replaces fragmented, informal coordination.
Data-Driven Decisions	Accurate revenue reporting lets owners optimize pricing and staffing.
Scalable, Secure Foundation	A robust technology stack capable of handling growth in bookings and properties.
Competitive Advantage	A superior digital booking and property-management experience versus manual competitors.

16. Conclusion

The Guesthouse Reservation Platform represents a significant advancement in digital accommodation booking for the Ethiopian market, addressing both the customer-facing booking experience and the operational reality of running a guesthouse. This revision simplifies the Version 1.0 six-role model into four accountable human roles — Guest, Guesthouse Owner, Receptionist, and System Administrator — supported by an automated System layer that performs authentication, availability management, payment verification, and notifications without manual intervention.

This document is intended to serve as the authoritative requirements baseline for design and development: it defines who uses the system, what each role is responsible for, what the System does automatically, what data the system must store, and what quality and security bar the implementation must meet. Future enhancements — guest reviews, a dedicated mobile application, multi-gateway payment integration, AI-powered recommendations, multi-language support, and deeper analytics — should be built on top of this simplified, automation-first foundation rather than around it.

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